

Characterization of Optical Camera Communication Based on a Comprehensive System Model

Anqi Liu¹, Wenxiao Shi¹, Min Ouyang², and Wei Liu¹

Abstract—Optical camera communication (OCC), as a complementary technology of radio frequency (RF), has played an important role in alleviating the burden on the existing spectrum and enhancing the reliability of communication. Theoretical investigations which describe the unique characteristics of camera-based optical wireless communication systems on a complete system model are limited, as most existing research utilized the indicators that are applied in photo detector (PD)-based systems. Due to the fact that cameras use pixel values to record the intensity of the LED, which carries the signals information, analysis based on pixel values has the ability to intuitively reflect OCC systems features. Motivated by this, a comprehensive system model which considers the photoelectric conversion and the imaging process of the camera is presented to characterize OCC systems in this paper. The proposed model which has been validated on the hardware platform describes the relationship between the LED driving current and the final pixel values on the image sensor. Some system parameters' impact has been analyzed based on the proposed system model and corresponding experimental results. Performance analysis has been provided to characterize OCC systems from the communication perspective. Our investigation may help introduce a new perspective to understand the principle of OCC, as well as providing the theoretical support needed for the design in practical systems.

Index Terms—Optical camera communication (OCC), pixel values, system modeling, system characteristics.

I. INTRODUCTION

THE pervasive connectivity of the Internet of Things (IoT) and the growth of digital services stimulate the rapid increase in the data being generated as well as the growing demand for capacity. According to the latest report of Ericsson, total mobile network traffic is forecast to exceed 300 exaBytes (EB) per month in 2026 [1]. Optical camera communication (OCC) which utilizes the huge license-free optical spectrum is an auspicious, green, and energy-efficient solution [2]. By taking advantage of the proliferating use of light-emitting diode (LED)

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as the transmitter and the cost-efficient off-the-shelf camera as the receiver, OCC has attracted great attention in academia, industry, military, and commercial fields since it was first included in the revised IEEE 802.15.7r1 standard in 2014 [3], and can be treated as a promising complementary technique of radio frequency (RF) to help realize a more reliable communication system [4], [5].

Improving communication performance of OCC systems is the goal many researchers pursue in this area. And a great many contributions have been made based on experimental tests, as well as system theoretical analysis [6]–[45].

For experimental tests, two categories of methods have been applied to improve the system performance, one is based on the modification of the hardware platform, and the other is based on effective but intelligent algorithms. For the first category of method, LED array [6], rotary LED [7], and RGB-LED [8] were applied to improve the transmission data rate of OCC systems. Delicately designed optical communication image sensor (OCI) [6], [9], high frame rate cameras with long focal length [6], and complementary metal-oxide-semiconductor (CMOS) cameras with rolling shutter characteristics [10]–[12] have been explored to promote the performance of practical OCC systems. For the algorithm-based solutions, different modulation methods [13]–[16] were proposed to improve the transmission efficiency of OCC systems, as well as realizing dimming control. A variety of coding mechanisms [17]–[19] were utilized to improve the transmission distance and enhance the anti-interference ability of the systems. Synchronization [20], equalization [21], and denoising [22] have been proposed to enhance the reliability of the systems as well. Machine learning-based algorithms, such as the application of Convolutional Neural Network (CNN) and other artificial neural networks [8], [23]–[25], have also played a significant role in realizing an effective but accurate demodulation process under various complicated surroundings.

As the performance of the experimental tests has a close relationship with the designers' hardware conditions and experimental environment, it's hard to apply the suggestions in generalized scenarios. Theoretical modeling, which is what we focus on in this paper, helps to provide guidelines for the generalized practical system design, and therefore becomes another approach to help improve the system performance. Many investigations have been made on the theoretical modeling, as well as the analysis of the parameters for LED transmitter, free space channel, and camera receiver to illustrate the characteristics of OCC systems [26]–[45].

For the transmitter LED, the basic properties of the LED lights have been demonstrated in [26]. Optimization of the LED Lambertian order [27], and the impact of the ambient light on the transmitter LED [28] were illustrated. Characteristics of the free space channel under fog and the atmospheric turbulence have been investigated in [29] and [30]. System modeling for line-of-sight (LOS) links, non-line-of-sight (NLOS) links, as well as scattering channels have been explored under different scenarios [31], [32]. For the camera receiver which can mostly reflect the differences among OCC and other optical wireless communication technologies, analysis based on the internal parameters which include but are not limited to the camera focal length, frame rate, aperture size, camera internal noises, resolution of the image sensor, analog gain, global-shutter (GS), as well as rolling-shutter (RS) mechanisms have been demonstrated in [33]–[37]. Imaging characteristics of the cameras have been presented from the photoelectric conversion perspective based on the received optical power and pixel values [38]–[42]. Frequency analysis for the camera receiver was demonstrated by treating it as an oscilloscope [43] and a linear shift-invariant receiver [44] respectively. Performance analysis for OCC through system modeling and the practical experimental results have also been made based on bit-error rate (BER), signal-to-noise ratio (SNR), peak signal-to-noise ratio (PSNR), and channel capacity [34], [37], [40], [45].

Although large quantities of contributions have been made in OCC research field, current studies still exist some limitations for theoretical analysis from the following perspectives. Firstly, many investigations only concentrated on a specific component of the OCC system [26], [31], [33]. As each part of the OCC contributes to the performance of the entire system, an overall characterization of OCC systems is in demand; Secondly, many studies for cameras description put emphasis on the spatial [33], rate [35], and frequency characteristics [43], [44] of the receivers, while a small proportion of them paid attention to their intensity characteristics [34], [37]–[39], [41]. As intensity-modulated (IM) /direct-detection (DD) method is applied in OCC field, knowing the system parameters' impact on the pixel values is intuitively helpful to the pixel value-threshold determination, which is beneficial to help realize a high-performance OCC system. [34] and [37] analyzed intensity characteristics of the camera through the imaging process based on optical power and photocurrent respectively. And these two indicators, which are often used in the analysis of photo detector (PD)-based optical wireless systems, are unable to directly reflect the characteristics of the camera imaging process, as cameras use pixel values to record the intensity of the received power. For the analysis based on pixel values [38], [39], [41], the research in [38] was based on the assumption that the radiated light intensity is a constant without considering the loss in actual communication systems, and the pixel values expressions presented in [39] and [41] were concept-based but not through rigorous verification. Therefore, a pixel value-based camera model which can be validated in practical systems is necessary for OCC field; Thirdly, performance analysis based on communication metrics is important to evaluate the communication potentiality of OCC systems. Many existing investigations analyzed BER and PSNR

based on the practical results [30]. However, few investigations explored the theoretically-described performance analysis on pixel value-based communication criteria, which can directly reflect OCC system characteristics considering camera as the receiver.

Motivated by this, in this paper, we introduce a comprehensive system model to analyze the characteristics of OCC systems based on the pixel values. The main contributions of this article can be summarized as follows.

- By jointly applying total transmitted optical power and light power distribution to describe the LED transmitter, modified channel direct current (DC) gain to express the free space channel, and mathematical formulas to depict the imaging process of the camera, we propose a comprehensive system model for OCC system to analyze its characteristics. Closed expression of the pixel value is obtained based on the proposed model which is suitable for both GS and RS-mode cameras.
- We analyze the parameters of the LED transmitter, free space channel, and camera receiver that may have an impact on the performance of the OCC systems under various scenarios respectively. The analyzed parameters include, but are not limited to the driving current of the LED transmitter, path loss coefficient of the channel, Gamma encoding value, as well as the internal noises of the camera.
- A virtual system has been set up on MATLAB to simulate the process of OCC communication systems based on the proposed system model. Through the simulation process, we provide a method to create relatively accurate captured LED images by adjusting simulated parameters according to different application scenarios.
- An OCC hardware platform has been set up to make comparisons with the simulation results. A relatively high fitting degree between the measured pixel values and the simulated data confirms the accuracy of our proposed system model.
- Performance analysis of PSNR, BER, and channel capacity has been investigated based on the pixel values obtained from our proposed model and the camera of our hardware platform.

The rest of this article is organized as follows. The proposed system model is introduced in Section II. Simulation implementation and hardware platform setup are described in Section III. Then in Section IV, discussions are made based on the related results. Finally, concluding remarks are presented in Section V.

Notations: $\odot$ represents Hadamard product, and if we define $C = A \odot B$, then the relationship of each element in the A , B , C matrix satisfies $c_{i,j} = a_{i,j}b_{i,j}$.

II. PROPOSED SYSTEM MODEL

In this section, we propose a comprehensive system model which includes the mathematical description of the LED transmitter, free space channel, and camera receiver. The influential parameters of each part of the proposed system model will be analyzed in detail. The schematic block diagram of the system model composition is shown in Fig. 1. The overall calculation

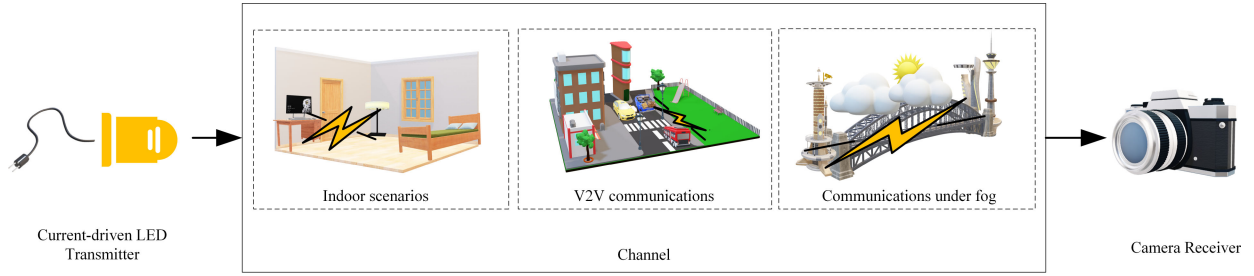


Fig. 1. The schematic block diagram of the system model composition, which includes the LED transmitter, free space channel, and camera receiver.

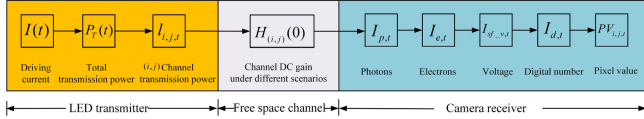


Fig. 2. The overall calculation process of the proposed system model.

process of our proposed system model is provided in Fig. 2, which demonstrates the relationship between the driving current of the LED and a certain pixel value of the camera image sensor.

A. LED Transmitter

1) *Power Characteristics of the LED Transmitter:* LED is a semiconductor light-emitting diode made of a P-N structure. By taking advantage of the spontaneous emission light generated when electrons are stimulated, it can provide illumination.

Optical transmission power is one of the basic properties of LED lights. According to [26], it can be calculated theoretically by integrating the energy flux within the range of the emission angle. As is often difficult to get the exact number of the energy flux, we use the driving current of the LED to describe its optical power. According to [46], the relationship between the driving current $I(t)$ and the transmitted optical power $P_T(t)$ can be given as

$$P_T(t) = kI(t), \quad (1)$$

where k is the linear coefficient factor determined by the physical properties of the LED. In OCC, as LEDs are used to transmit information, $I(t)$ and $P_T(t)$ vary with time according to the modulated signals. Here, we assume the transmitter LED is current-driven. Besides, the transmitter LED needs to be working under the linear region of the current-power relationship to avoid the signal distortion and performance degradation [47].

2) *Light Power Distribution of LED:* In the OCC field, camera receivers use the pixel values within the imaging area to record the states of the LED, as well as helping to demodulate the transmitted signals. Since the value of each pixel is related to the energy received on the corresponding area, it is not enough to treat the emitter as a point light source and only apply the total transmitted optical power to describe it.

Light power distribution, therefore, is proposed here to help describe the relationship between each part energy of the LED light source and the imaging pixel value accordingly. However, due to the imaging characteristics of the camera, it is only

meaningful to discuss the light power distribution of the LED within its imaging area. In our proposed model, for the sake of the subsequent calculations, we first use an $X \times Y$ array $\mathbf{S}$ to indicate the imaging area, where $X \times Y$ equals the resolution of the camera image sensor. According to the pinhole camera model [32], the element $s_{i,j}$ of row i and column j in $\mathbf{S}$ is given as

$$s_{i,j} = \begin{cases} 0, & |d_{i,j}| > R_i \\ 1, & |d_{i,j}| \leq R_i \end{cases} \quad (2)$$

where $|d_{i,j}|$ is the distance between the central point of $s_{i,j}$ and the central point of the imaging area $\mathbf{S}$. R_i is the imaging radius of the transmitter LED, and is given as

$$R_i = \frac{Rf}{D}, \quad (3)$$

where R is the physical radius of the assumed circular-shape transmitter LED, D is the transmission distance between LED and the camera, and f is the focal length of the camera. Here, considering the intuitiveness and simplicity of the pixel values distribution presentation, as well as highlighting our research emphasis, we only consider the situation that the LED and camera are parallel to each other, while ignoring the complex transformation process of the relevant coordinate systems.

Then, according to [48], we use the point spread function (PSF) to describe the light power spatial distribution of the LED. Besides, by operating PSF, fringes of light extending from the borders of an LED light source can also be modeled. Similar to [49], here we define an $X \times Y$ PSF array $\mathbf{P}$ with each element $p_{i,j}$ as

$$p_{i,j} = G(F)e^{\left(-\frac{d_{i,j}^2}{2\sigma^2}\right)}, \quad (4)$$

where σ^2 determines the degree of diffusion. $G(F)$ is the PSF gain. It's worthwhile to mention that other forms of PSF can also be used to describe the spatial distribution of stray light as in [50]. Since our proposed system model is given based on a single transmitter LED without considering the interference, PSF used in our system is only related to the transmitter itself.

Then, through (2) and (4), the light power distribution $\mathbf{W}$ within the actual imaging area can be given as

$$\mathbf{W} = \mathbf{S} \odot \mathbf{P}. \quad (5)$$

Finally, as the total transmission power of the LED is $P_T(t)$, based on the light power distribution given in (5), the transmission power of each area of the LED can be got from

$$\mathbf{L} = \frac{P_T(t)}{\sum_{(i,j) \in \mathbf{W}} w_{i,j}} \mathbf{W}, \quad (6)$$

where $l_{i,j,t}$ in $\mathbf{L}$ indicates the transmission power of a certain part of LED, whose imaging area is (i,j) on the image sensor.

B. Free Space Channel

In this subsection, a generalized optical channel model used for the description of the LOS channel DC gain in OCC is firstly given. Then according to different application scenarios, corresponding modified channel DC gain will be introduced. As we treat each pixel as an independent receiving area, namely, an independent channel, the above channel DC gains are proposed at a pixel level. Without loss of generality, we use subscript (i,j) to indicate such channel, whose imaging area corresponds to (i,j) pixel on the image sensor.

1) *Common LOS Optical Channel Model*: As the frequency responses of infrared channels are relatively flat near DC [51], we use channel DC gain $H_{los}(0)$ to characterize the LOS optical channel in the OCC field. For adapting to more generalized scenarios, we use the channel model which is slightly different from the one given in [51] as

$$H_{los(i,j)}(0) = \frac{A_p}{D_{i,j}^\gamma} R(\phi), \quad (7)$$

where $D_{i,j}$ is the distance between corresponding LED area and camera, γ is the optical path loss coefficient similar to [52], and $R(\phi)$ denotes LED radiant intensity. A_p is the signal collection area of the camera, and can be expressed as [51]

$$A_p = \begin{cases} AT_s g \cos(\theta_{i,j}), & 0 \leq \theta_{i,j} \leq \theta_l \\ 0, & \theta_{i,j} > \theta_l \end{cases} \quad (8)$$

where A is the area of the camera lens entrance pupil, T_s is the signal transmittance of the optical filter, $\theta_{i,j}$ is the angle of incidence (AoI), θ_l is the field of view (FoV) of the camera lens, and g is the lens gain. AoI in (8) allows the proposed system model to describe the intensity of the imaging LED over a range of deflection angles. Here, the incidence angles under which we can calculate the imaging LED pixel values are restricted to those who can guarantee that each part of the LED can be imaged under complete and non-overlapping conditions on the camera plane. Besides, in order to indicate the differences between common LOS channel links and NLOS links (NLOS links will be introduced in the indoor scenarios part), we put the schematic diagrams of the above two situations in Fig. 3. Description diagram of the common LOS channel model is shown in Fig. 3(a).

It is worthwhile to notice that although there are other LED patterns, i.e., piecewise Lambertian model [31], without loss of generality, we only consider the Lambertian-pattern LED in this paper. Based on this, $R(\phi)$ can be expressed as

$$R(\phi) = \frac{(m+1)}{2\pi} \cos^m(\phi), \quad (9)$$

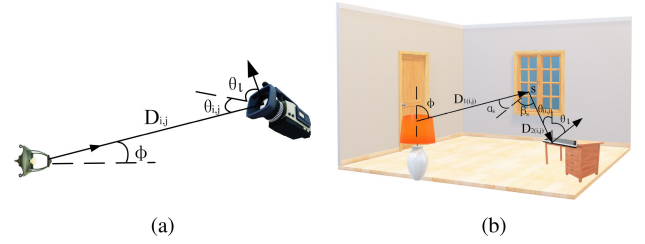


Fig. 3. The schematic diagram of the LOS and NLOS link. (a) A common LOS link. (b) An NLOS link in a typical indoor scenario.

where ϕ is the irradiance angle of the LED, m is the order of the Lambertian emission given as

$$m = \frac{-\ln(2)}{\ln(\cos(\Phi_{1/2}))}, \quad (10)$$

where $\Phi_{1/2}$ is the half-power angle of the LED source. Taking (9) and (8) into account, (7) can be re-written as

$$H_{los(i,j)}(0) = \begin{cases} \frac{(m+1)A}{2\pi D_{i,j}^\gamma} \cos^m(\phi) g T_s \cos(\theta_{i,j}), & 0 \leq \theta_{i,j} \leq \theta_l \\ 0, & \theta_{i,j} > \theta_l \end{cases} \quad (11)$$

where $H_{los(i,j)}(0)$ in $\mathbf{H}_{los}(\mathbf{0})$ is the (i,j) channel DC gain for the LOS link. And the received power of this channel can be given as

$$R_{los(i,j,t)} = l_{i,j,t} H_{los(i,j)}(0). \quad (12)$$

2) *Optical Channel Model Description Under Different Scenarios*: The modified optical channel model under indoor and outdoor conditions will be presented in this part.

a) *Indoor scenarios*: Under indoor scenarios, NLOS links need to be considered due to the existence of the surrounding walls as shown in Fig. 3(b). The received power from NLOS links can be calculated through the reflection processes from walls. As the first reflection from walls has the largest effect on the link performance, we provide the (i,j) channel DC gain $H_{ref(i,j,s)}(0)$ in $\mathbf{H}_{ref}(\mathbf{0})$ which only considers the first reflection similar to [53] as

$$H_{ref(i,j,s)}(0) = \begin{cases} \frac{(m+1)\Delta S \rho \cos(\alpha_s) \cos(\beta_s) \cos^m(\phi) g T_s \cos(\theta_{i,j})}{2(\pi D_{1(i,j)} D_{2(i,j)})^\gamma}, & 0 \leq \theta_{i,j} \leq \theta_l \\ 0, & \theta_{i,j} > \theta_l \end{cases} \quad (13)$$

where $D_{1(i,j)}$ is the distance between the LED and the first reflective point s , $D_{2(i,j)}$ is the distance between the first reflective point s and the receiver, ρ is the reflectance factor, α_s is the angle of incidence to the first reflective point, ΔS is the first reflection area, and β_s is the angle of irradiance to the receiver. α_s and β_s allow the proposed system model to calculate the received pixel values in the case of a certain range of deflection angles at the transceiver end.

Based on the above, the received power for channel (i,j) under the indoor scenarios can be expressed as

$$R_{indoor(i,j,t)} = R_{los(i,j,t)} + R_{ref(i,j,t)}, \quad (14)$$

where the second part of (14) means the power summation of the first-order reflections from all walls, and can be calculated

through

$$R_{ref}(i,j,t) = \sum_{all\ walls} \iint_S H_{ref}(i,j,s)(0) l_{i,j,t} dS, \quad (15)$$

where S represents the area of a single wall.

b) *Outdoor scenarios*: The increase of the transmission distance under the outdoor scenarios brings about the following two impacts: the first is the received pattern degradation and image blur, and the second is the received power attenuation and image distortion. Reduction of the pixel size, defocus of the LED, camera shake, as well as the high-speed movement of the transmitter are the main reasons leading to the first impact, whereas the second phenomenon is largely due to the weather influence.

According to [54], the vague pattern of the received image is due to the high spatial frequency loss over the transmission distance. And this phenomenon can be described by modeling the channel as a low-pass filter whose cut-off frequency varies according to the distance. Here, an $q \times q$ Gaussian filter $\mathbf{G}$ is used to model this phenomenon, and the convolution kernel $G_{a,b}$ of the filter is given as

$$G_{a,b} = \frac{1}{G_{sum}} \frac{e^{\left(-\frac{a^2+b^2}{2\sigma_G^2}\right)}}{2\pi\sigma_G^2}, \quad (16)$$

where (a,b) is the corresponding location coordinate of the convolution kernel, σ_G is the variance of the filter, G_{sum} is the sum of the convolution kernel weights and can be expressed as

$$G_{sum} = \sum_{a=-\frac{(q-1)}{2}}^{\frac{(q-1)}{2}} \sum_{b=-\frac{(q-1)}{2}}^{\frac{(q-1)}{2}} G_{a,b}. \quad (17)$$

Due to the introduction of the Gaussian filter, LOS channel DC gain considering the blur phenomenon can be given as

$$\mathbf{H}_{blur}(\mathbf{0}) = \mathbf{H}_{los}(\mathbf{0}) * \mathbf{G}, \quad (18)$$

where $*$ indicates the convolution operation, the element value of $\mathbf{H}_{los}(\mathbf{0})$ can be got from (11), and $H_{blur}(i,j)(0)$ in $\mathbf{H}_{blur}(\mathbf{0})$ is the (i,j) channel DC gain considering the blurry effect.

Based on the above, the received power of the camera under blurry situations can be given as

$$\mathbf{R}_{blur} = \mathbf{L} \odot \mathbf{H}_{blur}(\mathbf{0}), \quad (19)$$

where $R_{blur}(i,j,t)$ in $\mathbf{R}_{blur}$ is the received power of the (i,j) channel.

Meteorological phenomena, such as fog, rain, and snow will also cause the received image distortion, as well as the power attenuation in outdoor OCC systems [29]. In this paper, we take the analysis of fog as an example to explore its effect.

We use visibility V_m in km to describe fog attenuation A_f . According to Mie's scattering model [55], A_f can be given as

$$A_f = \frac{3.91}{V_m} \left(\frac{\lambda}{550} \right)^{-\delta}, \quad (20)$$

where λ is wavelength in nm, δ is the distribution size of the scattering particles given by Kim's model [56] as

$$\delta = \begin{cases} 1.6, & V_m > 50 \text{ km} \\ 1.3, & 6 \text{ km} < V_m < 50 \text{ km} \\ 0.16V_m + 0.34, & 1 \text{ km} < V_m < 6 \text{ km} \\ V_m - 0.5, & 0.5 \text{ km} < V_m < 1 \text{ km} \\ 0, & V_m < 0.5 \text{ km} \end{cases} \quad (21)$$

and then by applying Beer's law [57], the (i,j) channel coefficient for fog H_{fog} can be expressed as

$$H_{fog}(i,j) = e^{-A_f D_{i,j}}. \quad (22)$$

Then the (i,j) channel DC gain considering the fog effect is given as

$$H_{los_fog}(i,j) = H_{fog}(i,j) H_{los}(i,j)(0). \quad (23)$$

And finally, the received power of channel (i,j) under fog conditions can be expressed as

$$R_{fog}(i,j,t) = l_{i,j,t} H_{los_fog}(i,j). \quad (24)$$

In summary, if we use $R_{i,j,t}$ to indicate the received power by the (i,j) area of the camera image sensor, according to the above-mentioned scenarios, $R_{i,j,t}$ can be given as

$$R_{i,j,t} = \begin{cases} R_{los}(i,j,t) \\ R_{indoor}(i,j,t) \\ R_{blur}(i,j,t) \\ R_{fog}(i,j,t) \end{cases}. \quad (25)$$

C. Camera Receiver

The mathematical relationship between the received power and the pixel values of the image will be presented based on the camera receiver model in this subsection. Two different shutter modes of the camera (GS and RS-mode cameras) will also be considered in our proposed model. We will concentrate on the basic imaging process without considering the cameras' internal noises in the first part. In the second part, some common noises in the camera will first be introduced, after which, we will provide the modified description version of the imaging process to better simulate the actual scene. The block diagram of the camera imaging process is shown in Fig. 4.

1) *Basic Imaging Process Without Considering Camera Noises*: As the value of a certain pixel is related to the received power of the corresponding areas, we take a single pixel value calculation process as an example to describe the basic imaging process.

The received optical energy during the exposure time t_{exp} will be recorded in the form of photons $I_{p,t}$ first. For GS-mode cameras, all the pixels in an image frame are exposed simultaneously [5]. Therefore, GS cameras can only record the status of the LED transmitter at a single point of time per frame. The received photons for GS cameras can be calculated through

$$I_{p,t_GS} = \frac{\int_{t_s}^{t_s+t_{exp}} R_{i,j,t} dt}{Q_p}, \quad (26)$$

where t_s is the time point at which the exposure of a frame starts.

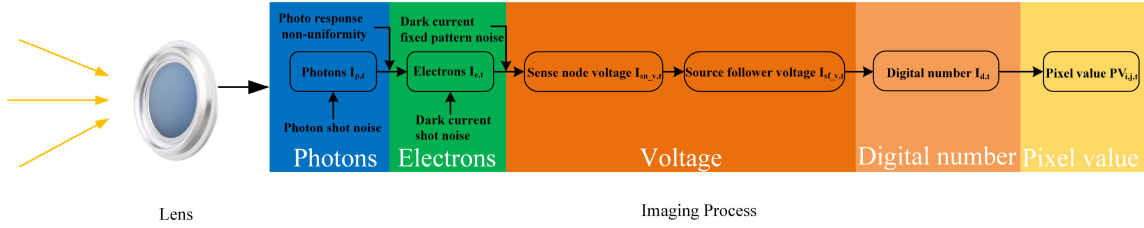


Fig. 4. Diagram of the camera imaging process, which describes the procedures from photons to final pixel values.

For RS cameras, as each row of the pixels is exposed sequentially, different rows of the RS cameras are exposed at different exposure time periods (known as the rolling-shutter feature) [11]. Due to this unique characteristic, RS cameras are able to record the different times status of the transmitter LEDs in one frame (certain frequency conditions of the LEDs need to be met) [10]. And RS cameras can help OCC systems to transmit with data rates that far exceed their frame rate. The received photons for RS cameras can be calculated as

$$I_{p,t_RS} = \frac{\int_{t_s+(i-1)t_H}^{t_s+(i-1)t_H+t_{exp}} R_{i,j,t} dt}{Q_p} \quad i \in X, \quad (27)$$

where t_s is also the time point at which the exposure of a frame starts, i is the row index number, and X is the vertical resolution (height) of the camera image sensor. The subsequent rows of RS cameras are exposed sequentially every t_H seconds. $R_{i,j,t}$ and Q_p in (26) and (27) are the received power of the (i, j) pixel and the energy of a single photon at wavelength λ respectively. $R_{i,j,t}$ can be got from (25), and Q_p can be calculated through

$$Q_p = \frac{hc}{\lambda}, \quad (28)$$

where h is Planck's constant and c is the speed of light.

The received photons $I_{p,t}$ according to different shutter modes can in summary be written as

$$I_{p,t} = \begin{cases} I_{p,t_GS} \\ I_{p,t_RS} \end{cases}. \quad (29)$$

The received photons $I_{p,t}$ are then transformed to electrons $I_{e,t}$ given as

$$I_{e,t} = I_{p,t} Q_e, \quad (30)$$

where Q_e is the quantum efficiency.

The electrons $I_{e,t}$ are then first converted to sense node voltages $I_{s,n,v,t}$ given as

$$I_{s,n,v,t} = V_{ref} - I_{e,t} A_n, \quad (31)$$

where V_{ref} is the reference voltage, A_n is the sense node gain. And then $I_{s,n,v,t}$ is transformed to source follower voltage $I_{s,f,v,t}$ as

$$I_{s,f,v,t} = A_f I_{s,n,v,t}, \quad (32)$$

where A_f is the source follower gain of the camera.

After the correlated double sampling (CDS) and analogue-to-digital converter (ADC) processes, $I_{s,f,v,t}$ is converted to the digital number as

$$I_{d,t} = A_a A_c (V_{a_ref} - I_{s,f,v,t}), \quad (33)$$

where V_{a_ref} is the maximum voltage that can be quantified by ADC, A_a and A_c are ADC gain and CDS gain respectively.

In order to get the final pixel value, according to [38], Gamma encoding is applied to the digital number given as

$$PV_{i,j,t} = 255 \times \left(\frac{I_{d,t}}{raw_{max}} \right)^{1/g_a}, \quad (34)$$

where raw_{max} is the maximum possible raw output value, and g_a is the gamma value. Here, we assume the pixel values range from 0-255.

Bring (26)–(33) into (34), the value of (i, j) pixel without considering the camera internal noises can finally be written as

$$PV_{i,j,t} = 255 \times \left\{ \frac{A_a A_c [V_{a_ref} - A_f (V_{ref} - Q_e A_n I_{p,t})]}{raw_{max}} \right\}^{1/g_a}. \quad (35)$$

2) *Modified Imaging Process Description With Some Camera Noises*: As limited by space, we only discuss the photon-based camera noises and the dark current-based camera noises in this part.

Photon-based noises can be roughly categorized as photon shot noise and photo response non-uniformity (PRNU). The former is due to the random arrival of photons, and can be modeled as a Poisson process P with mean $I_{p,t}$ as [58]

$$I_{p-s,t} = P(I_{p,t}), \quad (36)$$

where $I_{p-s,t}$ indicates the number of the received photons.

The latter is due to the spatial variation in the pixel output under uniform illumination, and can be simulated as a Gaussian distribution which obeys to $N(0, \sigma_{PRNU}^2)$ as described in [58].

Dark current is formed by the thermally generated electrons, and dark current-based noises can be roughly categorized as dark current shot noise and dark current fixed pattern noise (FPN). Similar to photon shot noise, the former can be modeled as a Poisson process P with mean I_{dc} given as

$$I_{dc} = t_{exp} D_r, \quad (37)$$

where D_r is the average dark current. And the latter can be simulated as a Log-Normal distribution which obeys to $\ln N(0, \sigma_{FPN}^2)$ [58].

Based on the above, the description of the imaging process needs to be modified. The received photons in (29) can be re-written as

$$I_{p,t} = I_{p-s,t}. \quad (38)$$

And the number of the total electrons can be calculated as [58]

$$I_{e,t} = I_{p,s,t} Q_e (1 + N(0, \sigma_{PRNU}^2)) + I_{dc} (1 + \ln N(0, \sigma_{FPN}^2)), \quad (39)$$

where the first part indicates the photon-generated electrons, and the second part represents the thermally generated ones.

After replacing (29) and (30) with (38) and (39) respectively, we get the new final pixel values under the existence of camera noises as (40) shown at the bottom of this page.

III. SIMULATION IMPLEMENTATION AND HARDWARE PLATFORM SETUP

Simulation experiments which include three processes are based on the proposed system model as shown in Fig. 5(a). According to the preset physical diameter, camera focal length, and transmission distance, the imaging radius of the LED can be calculated first as shown in Fig. 5(b). In our experiments, we only consider the condition that the LED and camera are parallel to each other. Then, with PSF and total transmission power, power allocation is conducted to corresponding areas of the LED. Based on the analysis scenarios, the light of the simulated LED transmits through different channel models, and is received by the camera. After the mathematically described imaging process, pixel values are got on the simulated image. Here, since we treat the individual pixel as an independent channel, with the help of the matrix operation function provided in MATLAB, all the pixel values in the scope of the imaging area can be calculated in parallel. Finally, through visualizing the distribution of the pixel values, the impact of the corresponding parameters on OCC systems can be explored.

The hardware platform we set up for the sake of making field tests are shown in Fig. 5(c). As Q_p in the proposed system model has a direct relationship with the wavelength of the light, the multi-wavelength LED light source may increase the difficulty when validating the accuracy of the proposed system model. Moreover, the interfering light source from the surroundings may also hinder our confirmatory experiments based on the hardware platform. In order to decrease the negative impact from the above two conditions, a monochrome LED array with 808 nm wavelength is applied as the transmitter, and a camera with 808 nm filter is used accordingly as the receiver. In order to provide the same experimental environment when studying different influential parameters of LEDs, i.e., driving current, we use an LED array instead of a single LED as the transmitter to enhance the experimental accuracy, and to reduce the time cycle of the experiments. The laser distance meter appeared in Fig. 5(c) is applied to measure the transmission distance in our experiments.

For validating purposes, we assume the LED driving current $I(t)$ is time-invariant under different conditions, and I is used to indicate it in the rest of our paper. Besides, a GS-mode camera is

utilized in both simulation and field-based experiments. Some of the key parameters remain the same which are shown in Table I. However, the parameters which may not have the impact on the final pixel values, i.e., LED physical radius, are not exactly the same for these two experiments, as limited by the hardware platform restriction and for the sake of a better display in the simulation diagram. Therefore, these parameters are not listed in Table I.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, we first aim to analyze the impact of the related parameters of the transmitter, channel, and camera receiver based on the simulated OCC systems. We also validate our proposed system model on our hardware platform under different conditions. Then performance analysis based on PSNR, BER, and channel capacity is presented through the acquired pixel values from our proposed model, as well as the tested ones from our hardware platform. As we treat each pixel as a channel, maximum luminance extraction [13] that can better reflect the influential degree of the corresponding parameters is applied in this paper.

A. Impact of the LED Transmitter

In this part, the impact of the LED driving current is presented.

LED driving current I influences the LED transmitted power directly according to (1). As the k value for our LED array transmitter is approximately 2 according to the practical measured results, for the sake of comparison and validation, we conduct the simulation experiments assuming k equals 1 and 2 respectively. Based on that, we firstly do simulations under different I from 35 mA to 95 mA. Then with the aid of a separate drive circuit for each LED on our hardware platform, seven driving currents equally divided between 35 mA and 95 mA are carefully allocated to 16 LEDs (the driving current of some LEDs may be the same). Therefore, we can rigorously investigate the driving current impact on the pixel values. One of the captured images of the LED array is shown in Fig. 6(a) at a transmission distance of 3 m. Simulation and field tests results are shown in Fig. 6(b).

From Fig. 6, we can notice that the simulation results fit well with the measured values under the variation of I . Simulation and field tests results also indicate that the improvement of k and I can both increase the pixel values. However, the linear increase of I brings the non-linear growing trend of the pixel values. Besides, the variation of the maximum pixel values becomes smaller with the increase of I . From later investigations, the Gamma encoding process in the camera is found as the main reason to lead this phenomenon. Detailed analysis will appear in the camera influence discussion part.

$$PV_{i,j,t} = 255 \times \left\{ \frac{A_a A_c [V_{a_ref} - A_f (V_{ref} - A_n (P(I_{p,t}) Q_e (1 + N(0, \sigma_{PRNU}^2)) + t_{exp} D_r (1 + \ln N(0, \sigma_{FPN}^2)))]}{raw_{max}} \right\}^{1/g_a}. \quad (40)$$

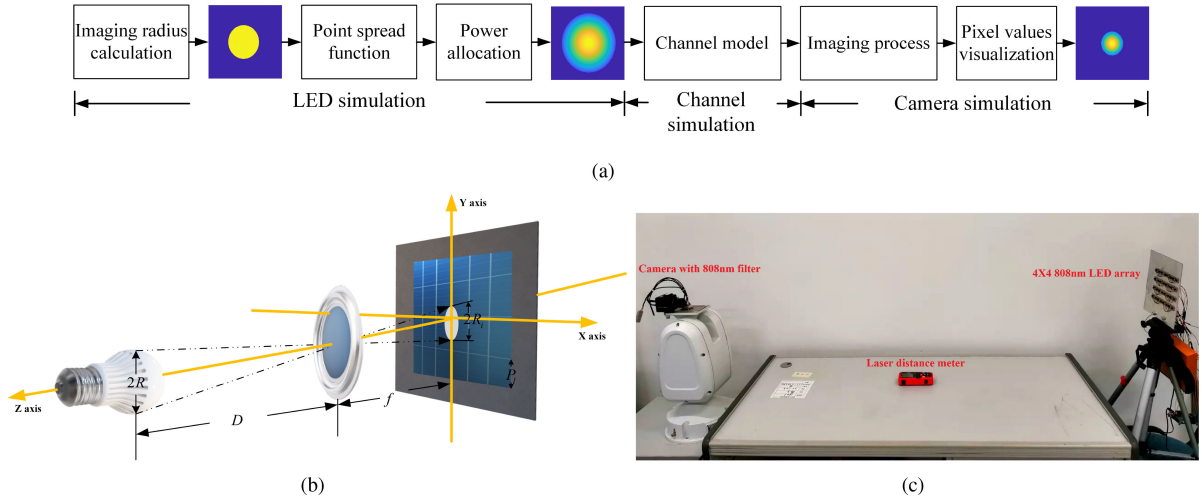
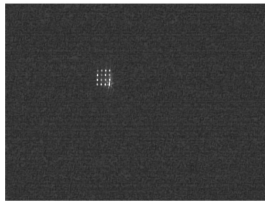


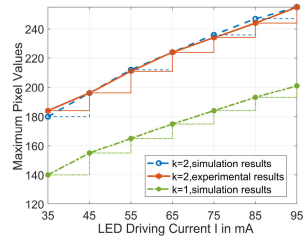
Fig. 5. Simulation procedures and field tests setup. (a) Simulation procedures. (b) Schematic diagram of the imaging radius calculation. (c) Field tests setup.

TABLE I
SIMULATION EXPERIMENTS AND FIELD TESTS PARAMETERS

Name of Parameter	Value of Parameter	Name of Parameter	Value of Parameter
LED irradiance angle, ϕ	20°	LED half-power angle, $\Phi_{1/2}$	18°
Angle of incidence, θ	0°	LED wavelength, λ	808 nm
Pixel length, P_l	$4.75 \mu\text{m}$	Camera resolution, $X \times Y$	720×576
FoV, θ_l	90°	Camera exposure time, t_{exp}	$1 \times 10^{-3} \text{ s}$
Quantum efficiency, Q_e	0.95	Reference voltage, V_{ref}	3.1V
Sense node gain, A_n	$5 \times 10^{-6} \text{ V/e}$	Source follower gain, A_f	1 V/V
ADC gain, A_a	1 V/V	CDS gain, A_c	1 V/V



(a)



(b)

Fig. 6. Maximum pixel values variation under different LED driving currents. Assume $D = 3 \text{ m}$, $f = 6 \text{ mm}$. (a) One of the captured images of the LED array. (b) Simulated and measured pixel values under different driving currents.

Moreover, it is worthwhile to mention that this non-linear increase may bring challenges to the design of high-order modulations which depend on methods in the OCC area. The carefully designed duty cycle of the pulse width modulation (PWM) in [13] is an example to indicate this fact.

B. Impact of the Free Space Channel

In this subsection, we first analyze the impact of the path loss coefficient γ which is often ignored in the existing research, and then make discussions on the proposed free space channel model under indoor and outdoor scenarios. As the impact of

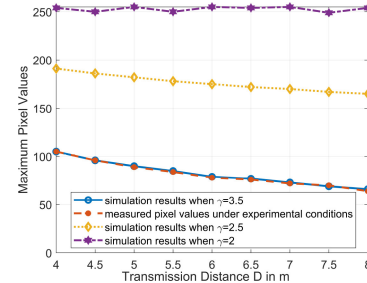


Fig. 7. The influences of different channel path loss coefficients. Assume $I = 35 \text{ mA}$, $f = 6 \text{ mm}$.

the transmission distance D has already been analyzed in our previous work [59], related discussions are ignored in this paper.

1) *Channel Path Loss Coefficient*: The values of γ we set for our experiments are 2, 2.5, and 3.5, where 2 is the most commonly used value in the OCC research field, and 2.5, as well as 3.5, are the measured ones from [31].

The simulation results shown in Fig. 7 indicate that when γ is 2.5 and 3.5, the increase of D will lead to the decrease of the pixel values. Besides, smaller γ corresponds to larger pixel values. However, when γ is 2, the pixel values hardly change with the variation of D . This interesting phenomenon coincides with the conclusion proposed in [32], and can be explained based on their pixel illumination model. When the imaging area is over

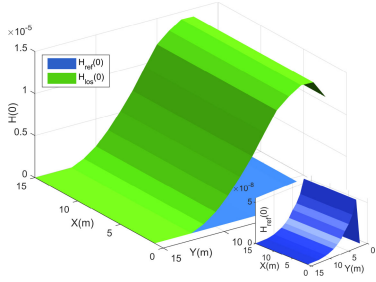


Fig. 8. Schematic diagram of the channel DC gain of LOS and NLOS links in indoor scenarios.

1 pixel (as D is small in our experiments, the imaging area is obviously larger than 1 pixel), the DC gain at a pixel remains constant, which then leads to a nearly constant pixel value. And the little variation is caused by the random noises in the camera.

We do field experiments based on our platform under different D values between 4m-8 m as shown in Fig. 7 as well. The results manifest that γ value of our environment is approximately 3.5. This conclusion reminds us that directly treating γ as 2 may not be reasonable, as a little variation of γ may lead to the obvious differences between the pixel values, which may bring a negative impact on the demodulation process.

2) *Indoor Scenarios:* In this subsection, we set up an indoor simulation environment to explore the impact of the received power from NLOS links on the OCC system. The size of the simulated environment is $5m \times 5m \times 2.15m$, which is the volume value of a typical office. As the first-order reflection from walls has the strongest energy in all NLOS links, we use this energy to represent power from all the possible NLOS links in our simulation. Besides, as the received power from the NLOS links is limited compared with those received from LOS links, in this experiment, we visualize the $H_{los}(0)$ and $H_{ref}(0)$ derived from (11) and (13) as shown in Fig. 8 to show the differences between them. The simulation results signify that the LOS channel DC gain is almost 1000 times to the NLOS channel DC gain in the simulated indoor environment.

These results indicate that the energy from LOS links plays the dominant role in the total received energy under our simulated environment. And the received power from NLOS links is not enough to increase the pixel values of the received images by one unit based on our model. Therefore, the NLOS links impact can be reasonably ignored when OCC is applied in a low-speed link with weak reflection and scattering conditions [5], [26]. However, when OCC is meant to be applied in high-data-rate situations where strong reflection or scattering phenomenon exists, inter-symbol interference (ISI) caused by the multi-path transmission should be carefully considered, and an equalization method may be needed [21].

3) *Outdoor Scenarios:* Two typical outdoor scenarios have been considered in this part which include a vehicle-to-vehicle (V2V) communication scene and the optical wireless environment under the fog condition.

For a V2V communication scene, motion blur caused by the high-speed movement of the vehicles is considered. We do the field tests by moving the LED array of our transmitter fast, and

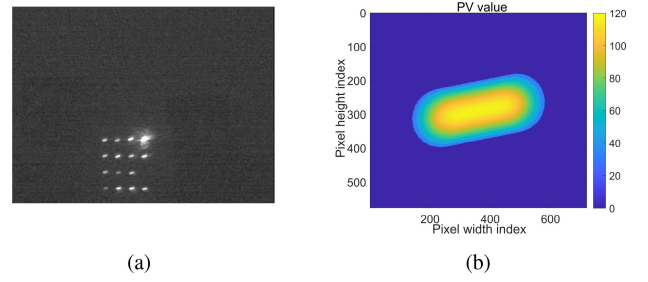


Fig. 9. Field test and simulation experiment which show the motion phenomenon. (a) One of the captured images when the LED array is moving, where $f = 6$ mm, $D = 3$ m. (b) Simulation of motion phenomenon, where $f = 35$ mm, $D = 3$ m.

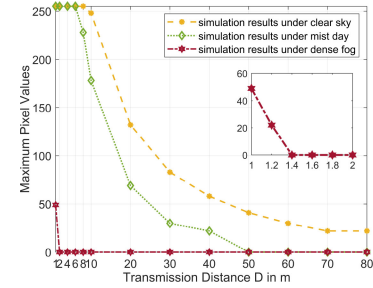


Fig. 10. The impact of different weather conditions on pixel values. Assume $f = 105$ mm, $I = 95$ mA.

keeping the camera receiver static. One of the captured images is shown in Fig. 9(a). In our simulation experiment, we apply the encapsulated 'motion filter' function in MATLAB to simulate the motion blur phenomenon as shown in Fig. 9(b). Here, we assume the movement length is 200 pixels on the simulated pictures. Similarities between the simulated motion phenomenon and the practically captured one validate the rationality of the application of the motion filter.

From Fig. 9, we can notice that the motion blur is obvious due to the large movement length, and the imaging LED has already been deformed. The deformed-imaging LED may provide great challenges to the detection method, deteriorate the OCC communication performance, and lead to a high BER. High vehicle speed and the relatively low frame rate of the cameras are the main reasons to cause motion blur.

We also make simulation experiments under three different weather conditions, which include simulations under the clear sky (visibility set as 50 km), mist (visibility set as 20 km), and dense fog (visibility set as 200 m). Besides, to show clearly the weather impact on the simulated OCC systems, we increase the transmission power of the LED and the camera focal length. Simulation results are shown in Fig. 10.

The simulation results indicate that the longest transmission distance is achieved under the clear sky condition, while the communication distance is sharply reduced under the dense fog condition. The denser the fog, the more obvious the attenuation of the pixel value caused by the fog under the unit distance. Since OCC is sensitive to the changes in weather, collaborative communication can be of great help to guarantee a reliable practical system under bad weather conditions.

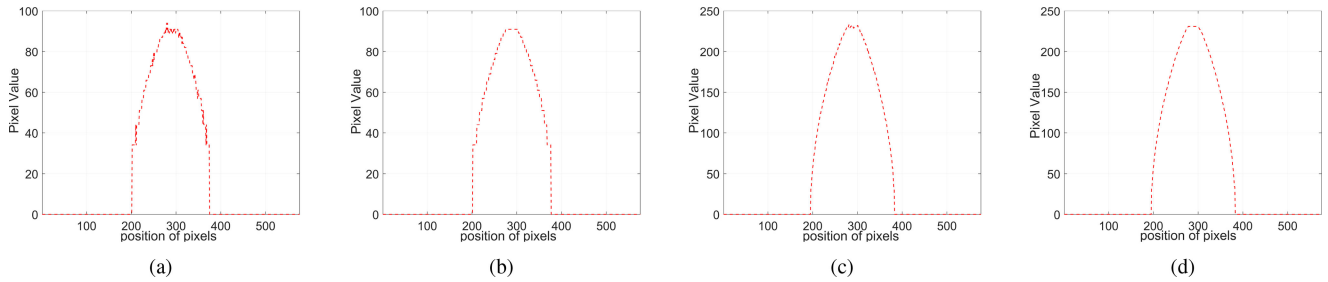


Fig. 11. The impact of the camera internal noises under different LED transmission power conditions. (a) Pixel values distribution with camera internal noises, while assuming $i = 10$ mA. (b) Pixel values distribution without camera internal noises, while assuming $i = 10$ mA. (c) Pixel values distribution with camera internal noises, while assuming $i = 90$ mA. (d) Pixel values distribution without camera internal noises, while assuming $i = 90$ mA.

C. Impact of the Camera Receiver

Many internal parameters of the camera may have an impact on the final pixel values of the received images. And the camera's internal noises, the imaging differences between the charge-coupled devices (CCD) and CMOS cameras, as well as the impact of the Gamma encoding process will be discussed in this subsection. As the influence from camera exposure time t_{exp} has already been analyzed in our previous research [59], related contents are ignored here.

1) *Camera Internal Noises*: Two main patterns of the camera noises which include photon-based noises and dark current-based noises are simulated in our experiments. For fixed-pattern (PRNU, FPN) noises, a fixed seed is allocated to each experiment to make sure that these noises are calculated once and then saved to a file. However, for the time-variant (photon shot noise, dark current shot noise) noises, random seeds are allocated to every experiment, and these noises are recalculated each time the simulation begins.

The impact of camera internal noises on the pixel values of the image has been investigated in our simulation experiments under the weak and strong transmission power conditions as shown in Fig. 11(a)–(b), and Fig. 11(c)–(d) respectively. Each sub figure visualizes the pixel values of the central column of the imaging LED in Fig. 11.

From the simulation results, we can notice that when the transmission power is weak, the pixel values on the two sides of the LED center point are unevenly distributed. The above phenomenon may limit the efficiency of the related LED detection methods which use the shape of the LED as the judging criterion. However, camera noises have little effect on the pixel values distribution when the transmission power is relatively high. Besides, the pixel values increment brought by the camera internal noises is small in both conditions.

2) *Differences Between CCD and CMOS Cameras*: CCD and CMOS are two main types of image sensors used in the OCC field.

For a CCD camera that only has one charge to voltage converter per sensor, it has to transfer each pixel's charge packet sequentially to convert its charge to a voltage, which limits its speed. Due to the lack of complicated structure in each pixel, a larger physical dimension can be achieved in a CCD camera which makes it more light-sensitive.

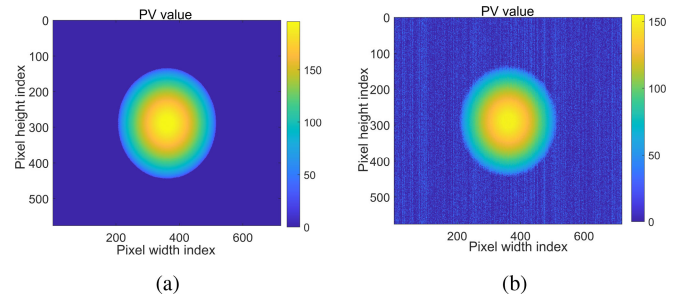


Fig. 12. Diagram of pixel values distribution for CCD and CMOS cameras. Assume $I = 45$ mA. (a) Schematic diagram of CCD camera imaging. (b) Schematic diagram of CMOS camera imaging.

For CMOS cameras, as each pixel is a discrete element hardwired to its own readout circuit, which means each pixel can be read individually and simultaneously, the read speed of the sensor is greatly enhanced. However, this fabricated structure also leads to variations in linearity and sensitivity between pixels, which greatly degrades the quality of the received images.

Many investigations in the OCC area directly applied CCD or CMOS cameras in their research without making performance comparisons of the two sensors. Apart from considering the photon-based camera noises and the dark current-based noises, more complex forms of the camera internal noises, i.e., sense node reset noise, as well as the offset fixed pattern noise, is brought in our simulation experiments which are inspired by the related work in [58] as well. However, different from [58], our research is made based on the visualization of the pixel values and analyzed from the OCC perspective. Simulation results are shown in Fig. 12.

Stripe-pattern noises on the entire image and severe burrs at the edge of the imaging LED can be noticed obviously on the CMOS-camera-captured image compared with the one captured by a CCD camera. The above phenomenon may greatly deteriorate the OCC communication performance if the high-clarity images are needed to support collaborative communication patterns with computer vision. However, as Intensity Modulated/Direct Detection (IM/DD) is often used in the OCC field, the intensity of the pixels in the imaging area of the LED is more important compared with the clarity of the images. Besides, considering the little impact on the imaging pixel values of the camera noises, CMOS cameras are preferable in most OCC research for their

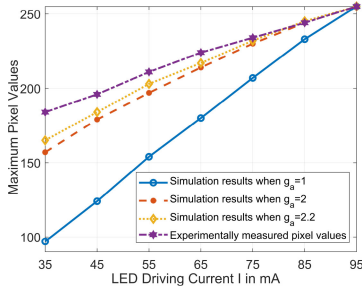


Fig. 13. Impact of Gamma value, where $f = 6$ mm, $D = 3$ m.

relatively cheap but fast scanning speed characteristics [5], [9], [10], [16].

3) *Gamma Encoding Process*: Gamma encoding process is proposed to compensate for the non-linearity of the human eye's perception of light. Few researchers in the OCC field paid special attention to this process in their paper. [38] only set the Gamma value as 2.2 which is common in the camera industry without providing a detailed analysis. Motivated by this, we study the Gamma encoding value's impact on the pixel values based on our proposed system model. For comparison, we set the Gamma value as 1, 2, and 2.2 respectively. We also provide the experimentally measured pixel values got from our practical system. Simulation and field test results are shown in Fig. 13.

From Fig. 13, we can notice that with the increase of the Gamma values, the differences between the minimum and maximum pixel values decrease. In practical OCC systems, these small differences may put a great challenge for the demodulation threshold estimation, as well as improving the complication of the related algorithms. Therefore, for the cameras whose Gamma values are adjustable, selecting an appropriate value may moderately broaden the differences between the maximum and minimum received pixel values, which may then lead to a high-performance OCC system at a much lower cost.

D. Performance Analysis

Performance analysis for OCC is useful in providing insights from a communication perspective. In this subsection, PSNR and BER are first presented based on the selected representative system parameters to characterize the quality of the OCC system, then discussions based on channel capacity are provided to describe the potentiality in the achievable transmission bit rate of OCC systems.

1) *PSNR Analysis*: As the camera is applied as the receiver in the OCC field, metrics for performance analysis not only need to be served as tools to measure the quality of OCC as a communication system, but also need to embody the unique characteristics of the camera. Due to the fact that the quality of the captured images has a close relationship with the accuracy of the demodulating process, according to [5], [30], and [60], PSNR is applied to evaluate the signal quality for OCC reception based on pixel values, which is given as

$$\text{PSNR} = 10 \log \frac{PV_{\max}^2}{\text{MSE}}, \quad (41)$$

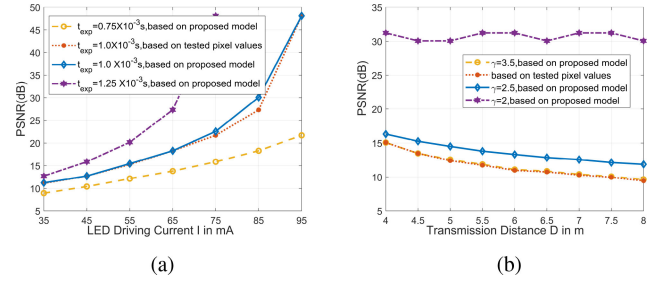


Fig. 14. Performance analysis based on PSNR under different LED driving currents I and different transmission distances D . (a) I -PSNR curve. (b) D -PSNR curve.

where $PV_{\max}$ is the maximum possible pixel value, and is 255 in our analysis. MSE is the pixel luminance mean squared error, and is given as

$$\text{MSE} = \frac{1}{z} \sum_{w=1}^z (PV_{Tx}(w) - PV_{Rx}(w))^2, \quad (42)$$

where PV_{Tx} are the average pixel values of the imaging area on the transmitter LED side, and similar to [30], we have obtained PV_{Tx} from captured or simulated images at approximately a zero distance from the transmitter. PV_{Rx} are the average pixel values of the imaging area on the camera receiver side, w is the image frame index number, and z is the number of frames.

As PSNR is based on pixel values, it helps to reflect the quantization limitations in cameras. We provide the I -PSNR and D -PSNR curve under three different t_{exp} and three different γ as shown in Fig. 14(a) and Fig. 14(b) respectively. And D corresponding to Fig. 14(a) is 3 m, I corresponding to Fig. 14(b) is 35 mA.

From I -PSNR curve, we can notice that when t_{exp} is fixed, PSNR increases with the improvement of I as long as no saturation phenomenon occurs. And with the increase of I , the I -PSNR curve becomes steeper. This is due to the fact that with the increase of I , pixel values variation becomes smaller (because of the Gamma encoding process), which then leads to the larger variation of PSNR. Fig. 14(a) also indicates longer t_{exp} results in higher values of PSNR under the same I , as long as there exists no saturation. This is obvious, as longer t_{exp} will enable the camera to receive more optical power. However, when saturation phenomenon occurs (this situation happens when t_{exp} is 1.25×10^{-3} s, and I is larger than 75 mA), pixel values won't change with the increase of the I , which means the camera receiver lacks the ability to reflect the intensity variation of the imaging area, and the communication functionality of OCC system is greatly deteriorated or even get lost. And PSNR can't provide useful suggestions under this situation, therefore is not presented in Fig. 14(a). We provide the results of PSNR based on the tested pixel values of our platform shown in Fig. 14(a) as well. And the variation trend of the experimental results is similar to the ones we got from our model.

From D -PSNR curve, we can notice that when γ is fixed and the value is not 2, PSNR decreases with the increase of D . However, when γ is 2, the PSNR nearly remains constant at a high level when we ignore the weather effect. As the pixel

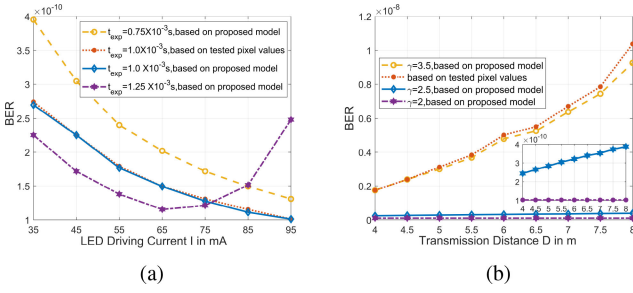


Fig. 15. Performance analysis based on BER under different LED driving currents I and different transmission distances D . (a) I -BER curve. (b) D -BER curve.

values hardly change with the variation of D (we have already presented and explained it in the impact of the LED transmitter part), the low-variation MSE caused by the random noises in the camera leads to a nearly constant PSNR at a high level when γ is 2. We also provide the results of PSNR based on the tested pixel values acquired from our platform shown in Fig. 14(b). And the variation trend of experimental results based on our OCC system under different distances coincides with the corresponding theoretical values got from our model.

2) *BER Analysis*: According to [61] and [62], when on-off-keying (OOK) is applied, BER (or P_e in some investigations [62]), which aims to illustrate the possibility of the wrong retrievals of the transmitted data from the received images in OCC systems can be calculated through

$$\text{BER} = \frac{1}{2} \text{erfc} \left(\sqrt{\frac{E_b}{2\sigma_n^2}} \right), \quad (43)$$

where pixel $\frac{E_b}{\sigma_n^2}$ can be calculated as $\frac{E_b}{\sigma_n^2} = \frac{E[s^2]}{E[n^2]}$ [61], and s is the pixel value which can be got either from our proposed model or from the images captured by our camera, and $\text{erfc}(\bullet)$ is the complementary error function. As the pixel noises in cameras can be approximately modeled as $n \sim N(0, \sigma(s)^2)$ [62], the variance of the noises can be calculated based on the variance of the received pixel values. As we use the maximum luminance extraction method, for simulation experiments, we simulate 500 pictures under each condition and calculated the variance of the noises based on the maximum pixel value of each picture. For field tests, we capture 500 pictures under each condition and do exactly the same calculation process as the simulation experiments.

Based on the results, I -BER curve under three different t_{exp} and D -BER curve under three different γ are presented in Fig. 15(a) and Fig. 15(b) respectively.

For I -BER curve, we can notice the increase of either I or t_{exp} will lead to the decrease of BER when no saturation phenomenon occurs. However, when saturation occurs (when t_{exp} is 1.25×10^{-3} s, and I is larger than 75 mA), E_b in (43) reaches the peak and does not increase with the improvement of the received power, while σ_n^2 in (43) still increases, which then leads to the increase of BER. For D -BER curve, Fig. 15(b) indicates that when γ is not 2, either the increase of γ or D will cause the increase of BER. However, when γ is 2, as the pixel

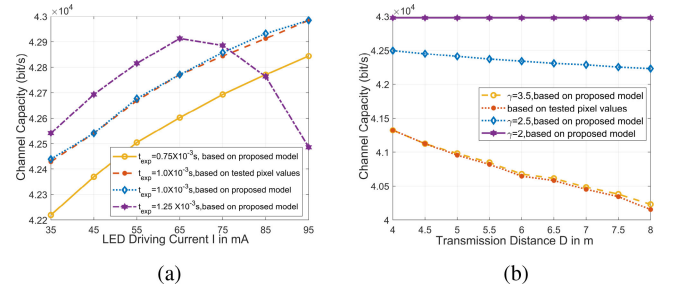


Fig. 16. Performance analysis based on C_{OCC} . (a) I - C_{OCC} curve. (b) D - C_{OCC} curve.

values hardly change with the variation of D (we've mentioned in PSNR analysis), the low-variation of the camera random noises lead to a nearly constant BER. And the variation trend of the calculated BER based on the tested pixel values from our captured pictures coincides with the calculated results got from our proposed model as shown in Fig. 15.

3) *Channel Capacity Analysis*: Similar to [40] and [45], the channel capacity C_{OCC} in bit/s of an OCC system can be calculated through

$$C_{OCC} = W_{fps}(W_s \log_2(1 + \text{SINR})), \quad (44)$$

where W_{fps} is the camera frame rate, and W_s is the spatial bandwidth, which is equivalent to the number of orthogonal or parallel channels in a Multiple-Input-Multiple-Output (MIMO) system. In our experiments, the maximum frame rate of our camera is 500 frames per second (fps). We use a 4×4 LED array as the transmitter, and W_s is 16 in our system. As we only consider the situation that the LED and the camera are parallel to each other, SINR can be given as $\text{SINR} = \frac{P_{avg}}{\sigma_n^2}$ [45], where P_{avg} denotes the average transmit pixel intensity, and we got the value of P_{avg} through calculating the average pixel values within the LED imaging field based on either the simulated pictures according to our model or the practically captured ones. We acquire the variance of noises through the same method introduced in BER analysis.

I - C_{OCC} curve under three different t_{exp} and D - C_{OCC} curve under three different γ are shown in Fig. 16(a) and Fig. 16(b) respectively. And D corresponding to Fig. 16(a) is 3 m, I corresponding to Fig. 16(b) is 35 mA.

For I - C_{OCC} curve, Fig. 16(a) denotes that when t_{exp} is fixed, the increase of I brings the increase of C_{OCC} when no saturation phenomenon occurs. However, when saturation happens, C_{OCC} decreases a lot (when t_{exp} is 1.25×10^{-3} s, and I is larger than 75 mA). To avoid confusion, the impact of t_{exp} on C_{OCC} is analyzed in two cases. For a clear analysis, the sampling time period, which is also the inverse of the camera frame rate (we only discuss GS cameras in our analysis) is described first. The sampling time period consists of three components, which include the reset time period, exposure time period, and the readout time period [5]. The first case is based on the condition that we increase t_{exp} while keeping the sampling time period remains fixed. The fixed sampling time period leads to a fixed value of W_{fps} in (44), and the increase of SINR caused by the improvement of t_{exp} brings the increase of C_{OCC} as shown in

Fig. 16(a) when no saturation occurs. A similar conclusion is also presented in [34], [37], and [44] described from the camera frequency perspective. The second case is based on the condition that the increase of t_{exp} leads to the increase of the sampling time period. As the improvement of the sampling time period means the decrease of W_{fps} in (44), the variation trend of C_{OCC} depends on the parameter between W_{fps} and SINR that has a larger influence. For example, if the increase of t_{exp} leads to a significant decrease of W_{fps} , though SINR increases, C_{OCC} still decreases. Similar analysis has also been mentioned in [42] and [61]. Besides, through (44), increasing the camera frame rate W_{fps} is also a direct way to improve C_{OCC} .

For D - C_{OCC} curve, Fig. 16(b) manifests that the increase of D , as well as γ (when γ is not 2) leads to the decrease of C_{OCC} . Due to the interesting properties of the pixel values when γ is 2, C_{OCC} nearly remains as a constant. And the variation trend of the calculated C_{OCC} based on the tested pixel values from our captured pictures coincides with the calculated results got from our proposed model as shown in Fig. 16.

V. CONCLUSION

In this paper, we have dedicated ourselves to handling the issue of characterizing OCC performance based on the pixel values. Simulation and field tests results indicate that the driving current of the transmitter LED, path loss coefficient of the space channel, as well as the weather, have a significant influence on the pixel values, whereas the impact from camera internal noises and the first-order reflection from the walls of a typical office is of little concern. High-speed cameras and efficient tracking algorithms are helpful to guarantee the reliability of the practical fast speed-based OCC systems. Fast-scanning speed, together with the cost-efficient price makes the CMOS camera a good choice as the receiver. Besides, selecting an appropriate Gamma value may lead to a high-performance OCC system at a much lower cost. Moreover, the impact of some representative system parameters based on PSNR, BER, and channel capacity has also been illustrated.

Analysis based on the condition that the transceiver ends are parallel to each other may lay the foundation for future related theoretical system model investigations, and be beneficial to the practical system parameters design and demodulation threshold estimation under various application scenarios. Due to the limited mobility support of our proposed system model, a more universal expression that considers multiple LEDs under generalized incidence angles as the transmitter will be investigated in the future for OCC systems.

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